Confidential

MarketingPilot AI Technical Proposal

Professional software development report for client review, investor sharing, and implementation planning

This proposal describes the full technical shape of the MarketingPilot AI platform, including product scope, technology stack, architecture, infrastructure, deployment path, services, cost structure, and post-launch operations.

The starter phase is planned as a lean launch package at **INR 10,000 to INR 15,000**. The goal is to build a clean, credible foundation first and expand into the full SaaS in later phases.

Website: https://progvision.online/ | Support: support@progvision.online

Client	MarketingPilot AI / ProgVision
Objective	Centralized AI marketing platform with content, CRM, WhatsApp, email, SEO, ads, and automation
Prepared by	ProgVision
Document control	Version 1.0 | Date: 21 July 2026 | Confidential

Table of Contents

Section	Page
Executive Summary	3
Technology Stack	4
System Architecture	5
Development Scope	6
Infrastructure & Deployment	7
Third-Party Services	8
Development Cost	9
Post-Launch Running Cost	10
Deployment Process	11
Maintenance & Support	12
Timeline	13
Assumptions & Exclusions	14
Conclusion	15



Executive Summary

This section introduces the platform, the target launch direction, and the deliverables that define the starter release.

1. Executive Summary

MarketingPilot AI is positioned as a centralized AI marketing and automation platform that combines content creation, lead handling, CRM actions, email workflows, WhatsApp workflows, SEO support, ad assistance, and business automation in one product.

The starter phase creates a polished and trustworthy foundation. It covers the first visible product shell, brand presentation, core workflows, and the infrastructure needed to prove the concept before the full production version is funded.

Item	Summary
Project overview	All-in-one SaaS for marketing operations, lead nurturing, and AI-assisted execution.
Objectives	Launch a credible starter release, validate the platform, and prepare for scale.
Scope	Landing page, login, admin shell, lead import, template library, integrations, and starter AI workflows.
Deliverables	Technical report, system architecture, infrastructure plan, cost sheet, timeline, and launch assumptions.

2. Technology Stack

Layer	Technology	Purpose	Why selected
Frontend	▲ Next.js 15 ⚛️ React 19 🟦 TypeScript 5 🎨 Tailwind CSS 4 🧩 shadcn/ui ✨ Framer Motion	Server-rendered UI, routing, dashboard screens, and interactions	Modern, fast, SEO-friendly, and suitable for enterprise-grade web apps
Backend	🟧 Node.js runtime 🧠 FastAPI / Express 🔁 REST API ⚙️ Background workers	Business logic, request handling, workflows, queues, and async jobs	Flexible enough for API-first SaaS development and automation
Database & Auth	🗄️ PocketBase 🔐 Authentication layer 📁 File storage	User accounts, session handling, records, and uploads	Lightweight for the starter phase and easy to operate at low cost
AI & Automation	🤖 Vercel AI SDK 🌐 OpenRouter free models 🧠 DeepSeek-R1 🧠 Qwen 🧠 Llama 3.1 📚 RAG engine	Inference, content generation, business analysis, and knowledge retrieval	Supports low-cost AI usage while keeping provider flexibility
Cloud & Deployment	🐧 Ubuntu 24.04 LTS 🌐 Nginx ☁️ Cloudflare 🔒 Let's Encrypt 🚀 GitHub Actions 🧭 PM2	Server hosting, reverse proxy, SSL, CI/CD, and process management	Stable, standard, and widely used for production SaaS deployments
Email & Integrations	✉️ Zoho Mail / SMTP 💬 WhatsApp API 📣 Meta Ads 🔎 Google Ads 🔗 Webhooks	Invites, alerts, campaign delivery, and external system connections	Necessary for real client communication and operational automation

React 19

Purpose: interactive user interface

Reusable components and responsive screens for the app experience.

Version: 19

▲ Next.js 15

Purpose: SSR, routing, SEO

Supports server-rendered pages, performance, and public SaaS visibility.

Version: 15

TypeScript 5

Purpose: type-safe development

Reduces bugs and keeps the codebase maintainable as the product grows.

Version: 5

Tailwind CSS 4

Purpose: responsive styling system

Consistent UI with fast iteration and clean layout control.

Version: 4

PocketBase

Purpose: auth, database, file storage

Keeps the starter infrastructure lightweight and cost-effective.

Version: latest stable

OpenRouter free models

Purpose: low-cost AI inference

Launches with free or low-cost AI before moving to paid models.

Models: DeepSeek-R1, Qwen, Llama 3.1

Cloudflare

Purpose: DNS, SSL, CDN

Improves security, caching, and delivery reliability at low cost.

Tier: free / paid as needed

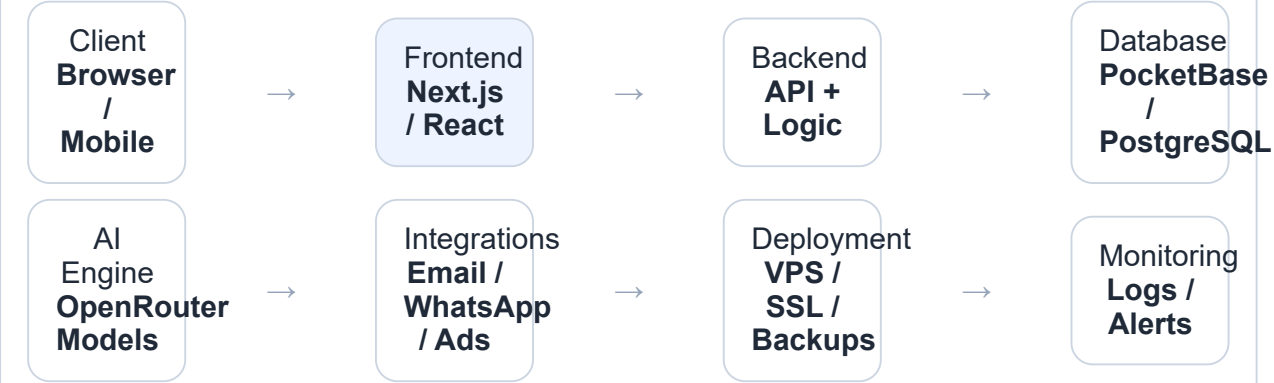
Ubuntu 24.04 LTS

Purpose: production server OS

Stable long-term Linux base for running the VPS and deployment services.

Version: 24.04 LTS

3. System Architecture



Architecture flow: the client uses the frontend, the frontend communicates with backend APIs, the backend reads and writes to the database, AI services support generation and analysis, and integrations connect the app to email and external channels. Deployment is managed on a VPS with SSL, backups, and monitoring.

4. Development Scope

Module	Features	Deliverables	Effort
Landing page	Hero, positioning, CTA, trust signals	Public marketing entry point	Low
Login and admin	Auth screens, role control, session handling	Secure access shell	Low
Lead import	CSV, Excel, manual rows, validation	Lead intake interface	Medium
CRM and workflow	Lead status, follow-up, tasks, notes	Starter CRM layer	Medium
Templates	Centralized email and WhatsApp templates	Message library + preview	Medium
AI modules	Content, SEO, ads, scripts, suggestions	AI-assisted workbench	Medium
Integrations	SMTP, WhatsApp, Meta, Google, webhooks	Integration setup pages	High

Infrastructure & Deployment

The next section shows where the starter version runs, how the server is configured, and what monthly operational cost to expect after deployment.

5. Infrastructure & Deployment

Area	Plan	Notes
Hosting	Small VPS or cloud VM	App server, API, and runtime services in one controlled environment.
VPS specification	2 vCPU / 4 GB RAM to start	Enough for a lean starter release; upgrade if traffic grows.
Domain and SSL	Custom domain + Cloudflare / Let's Encrypt SSL	Required for trust, security, and production use.
Database	PocketBase starter database	Simple to deploy now; can later move to PostgreSQL if needed.
Backup strategy	Automated daily backup job	Keep database, uploads, and config backups off-server.
CI/CD pipeline	Manual first, automated later	Git-based deploy flow can be added after the pilot stage.
Monitoring	Logs + uptime alerts	Track app health, mail failures, and API issues.
Security	HTTPS, access control, secrets management, rate limiting	Basic production safeguards for a public SaaS launch.

6. Third-Party Services

Service	Purpose	Free / Paid	Monthly cost	Required?
PocketBase	Starter database, auth, and file storage	Free	INR 0	Yes
OpenRouter	Free AI model access for generation	Free tier	INR 0 - INR 2,000	Yes
Zoho Mail / SMTP	Business email, invitations, alerts	Paid or bundled	INR 500 - INR 2,000	Yes

Cloudflare	DNS, SSL, caching, protection	Free / Paid	INR 0 - INR 1,500	Recommended
VPS Hosting	Runs frontend, backend, and services	Paid	INR 1,500 - INR 5,000	Yes
Domain	Public site and app identity	Paid	INR 80 - INR 200 monthly equivalent	Yes
Monitoring tools	Uptime and log review	Free / Paid	INR 500 - INR 2,000	Recommended

7. Development Cost

Area	Estimated cost
UI/UX	INR 2,000 - INR 3,000
Backend	INR 2,000 - INR 3,000
AI Engine	INR 1,500 - INR 2,500
CRM	INR 1,000 - INR 2,000
Automation	INR 1,500 - INR 2,500
Testing	INR 1,000 - INR 1,500
Deployment	INR 1,000 - INR 1,500
Documentation	INR 500 - INR 1,000

Starter development budget

INR 10,000 - INR 15,000

This budget covers the starter release only. Larger production scope will require a separate estimate.

8. Post-Launch Running Cost

Cost item	Expected monthly cost	Comment
Hosting	INR 1,500 - INR 5,000	Server, app runtime, and storage.
Database	INR 0 - INR 2,000	PocketBase starter or managed DB later.
AI usage	INR 0 - INR 2,000	Free model usage plus growth margin.
Email service	INR 500 - INR 2,000	SMTP or Zoho Mail for production mail.
Domain renewal	INR 80 - INR 200 monthly equivalent	Annual renewal averaged monthly.
Monitoring and backup	INR 500 - INR 2,000	Logs, uptime checks, backups, alerts.
Maintenance	INR 1,000 - INR 3,000	Bug fixes, support, and minor updates.

9. Deployment Process

Phase 1
Develop the UI, backend, and core workflows locally.

Phase 2
Connect database, email, AI, and integration services.

Phase 3
Deploy to VPS, enable SSL, configure domain and DNS.

Phase 4
Run testing, monitoring, backups, and production checks.

10. Maintenance & Support

Plan	Coverage	Typical monthly cost
Basic support	Bug fixes, small updates, and health checks	INR 1,000 - INR 3,000
Standard support	Monitoring, backups, fixes, and feature touch-ups	INR 3,000 - INR 7,000
AMC / growth support	Ongoing enhancements, new modules, and operational follow-up	Custom

11. Timeline

Phase	Duration	Outcome
Phase 1	2 - 3 days	Scope, branding, layout, and technical planning
Phase 2	4 - 6 days	UI, login, dashboard, lead import, and template shells
Phase 3	3 - 5 days	Integrations, AI wiring, deployment, and tests
Phase 4	1 - 2 days	Polish, handover, and report finalization

12. Assumptions & Exclusions

Included	Optional / not included yet
Starter SaaS shell, base workflows, infrastructure planning, technical report, and launch-ready presentation.	High-scale AI usage, advanced WhatsApp automation, heavy paid ads integration, large production infrastructure, and custom enterprise SLAs.

13. Conclusion

MarketingPilot AI can be launched as a polished starter product within a lean budget if the first release stays focused on the core user journey and the infrastructure remains intentionally lightweight. The platform is designed to scale later into a complete AI marketing suite with CRM, automation, ads, SEO, and communication workflows.

This report gives a client or investor clear visibility into the architecture, technology stack, deployment path, operational cost, and growth roadmap.